Rank spectra for the Coxeter source

Human Compression project: F006 recovery

Assembly for review

This manuscript joins three complete components concerning the principal results of Chen and Rowley, *Abstract Regular Polytopes of Finite Irreducible Coxeter Groups*, Electronic Journal of Combinatorics 32(4) (2025), P4.18. We use the fixed thirteen-page published edition. The source's definition permits a string diagram to be disconnected; its discussion of B_n explicitly uses that convention. All statements below retain it unless connectedness is specified.

Theorem 1. *For $n \geq 5$, the group $W(D_n)$ admits string C-markings of every rank from three to n when n is odd, and from three to $n-1$ when n is even. Those upper endpoints are its maximum ranks. For the exceptional groups, the maximum ranks are*

W	H_3	H_4	F_4	E_6	E_7	E_8
$r_{\max}(W)$	4	4	4	5	7	7
maximum connected rank	3	4	4	5	6	7.

For $I_2(m)$, $m \geq 3$, the maximum rank is three when $m \equiv 2 \pmod 4$ and two otherwise. Its connected maximum is always two. The six connected exceptional censuses agree with every entry of the source's Table 1. With disconnected strings included, H_3 has three additional classes at rank four, and E_7 has $462, 276, 72, 18$ additional classes at ranks four, five, six and seven, respectively, all modulo isomorphism and duality. None of these additional classes is self-dual.

Proof. The acquired *Even sign changes and the rank of a string* proves the upper bounds for $W(D_n)$ through its even sign module and the independent generating families in the symmetric quotient. Its companion *Constructing the rank interval with sign changes* proves the displayed constructions and their full intersection property. In odd degree, the acquired contraction theorem gives every intermediate rank down to three; in even degree, the explicit families give rank three and each rank from four to $n-1$. These statements apply to all the specified degrees, with no finite-degree extrapolation.

The acquired *Disconnected strings in two Coxeter families* proves the uniform dihedral assertion, the explicit H_3 rank-four witness, and the complete small census. Theorem 1 of *The exceptional Coxeter censuses*, with its full enumeration proofs and retained executions, gives the other five groups. Their final nonzero rows give both kinds of maximum. The separate direct subset-intersection verification also supplies an explicit disconnected rank-seven E_7 witness. Subtracting the connected E_7 rows from its complete rows gives $1987 - 1525 = 462$, $741 - 465 = 276$, $167 - 95 = 72$ and $18 - 0 = 18$. The self-dual columns agree in each subtraction. The actual comparison with the source's six printed rows is retained, including all zero entries and the total columns. □

The exact recovery contract. The first assertion recovers source Theorems 1 and 2, including both the upper bounds and all intermediate ranks. The remaining assertions recover Theorem 3 and Table 1 with their disconnected scope corrected in precisely $I_2(m)$, H_3 and E_7. The separate D_n source map records every intermediate numbered statement and retained input: in particular, its new kernel calculation corrects a displayed intermediate intersection in

Lemma 19 while preserving that lemma's rank-three conclusion. These are corrections to fixed published assertions, not answers to open conjectures.

The source's constructions, conclusions and connected counts remain attributed to Chen and Rowley. The new arguments are the explicit even-module and kernel derivations, the treatment of disconnected strings, and the complete new finite executions. Classical Coxeter identifications and orders, the cited symmetric-group generation results, the string end criterion, and the acquired rank-contraction theorem remain stated inputs. The enumeration follows the established Hartley–Hulpke and Leemans–Mixer method with the exact implementations and their proofs retained. No result about the cited external literature as a whole, historical priority, or other editions is asserted.

The proofs depend on the specified exact Python and GAP computations where explicitly stated. Those programs are computational premises and checks; they are not complete Lean formalizations. This assembly introduces no compression measurement and does not turn the source's corrected assertions into claimed literal recoveries of their false unrestricted form.